

# Regime-Adaptive Multi-Instrument Trading Strategy

Walk-forward ML signal · Curve regime classifier · 7-instrument portfolio · OOS 2021–2026

**Products:** CL (WTI) · LCO (Brent)    **OOS period:** Jan 2021 – Jun 2026    **Return horizon:** 20-day forward

**Signal:** Elastic Net + XGBoost ensemble    **Date:** July 2026

## PROJECT PHASES

|                                     |                                    |                                |                            |                                        |
|-------------------------------------|------------------------------------|--------------------------------|----------------------------|----------------------------------------|
| Phase 1                             | Phase 2                            | Phase 2B                       | Phase 3                    | Phase 3C                               |
| <b>Data Pipeline</b>                | <b>ML Signal</b>                   | <b>Regime Classifier</b>       | <b>Instrument Analysis</b> | <b>Strategy ✓</b>                      |
| EIA · FRED · freight · curve · CFTC | Walk-forward ensemble<br>OOS 2021+ | 9 regimes from curve structure | 7 instruments × 9 regimes  | Adaptive weights · direction-corrected |

## HEADLINE RESULTS — OOS 2021–2026

|                                                                |                                                            |                                                                |                                                              |                                                                  |                                                                 |
|----------------------------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>PORTFOLIO SHARPE</b><br><b>11.4</b><br>vs 5.8 outright-only | <b>MAX DRAWDOWN</b><br><b>–13.8%</b><br>vs –65.3% outright | <b>HIT RATE</b><br><b>82.5%</b><br>portfolio · 400 active days | <b>CALMAR</b><br><b>0.82</b><br>vs 0.09 outright (9× better) | <b>INSTRUMENTS</b><br><b>7</b><br>outright · 4 spreads · 2 flies | <b>REGIMES</b><br><b>9</b><br>back · cont · stable · transition |
|----------------------------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------|------------------------------------------------------------------|-----------------------------------------------------------------|

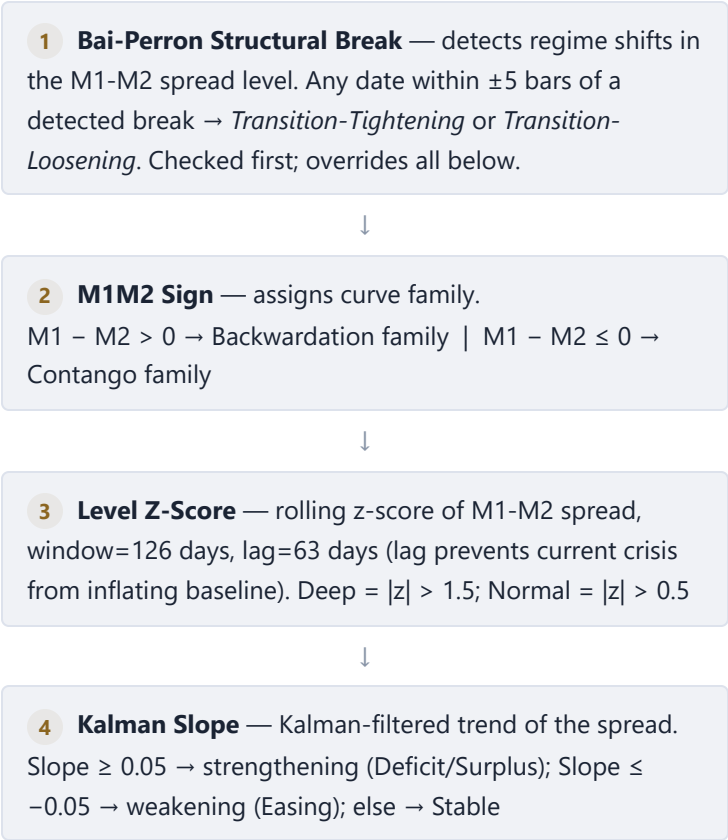
## DATA SOURCES

| Price & Curve Data                                                                      |  |
|-----------------------------------------------------------------------------------------|--|
| WTI (CL) outrightsgroup="1" data-cs="2" data-kind="parent">>M1–M12 · daily · 2019–2026  |  |
| Brent (LCO)group="1" data-cs="2" data-kind="parent">>M1–M12 · daily                     |  |
| Calendar spreadsgroup="1" data-cs="2" data-kind="parent">>M1–M2, M1–M3, M1–M6, M1–M12   |  |
| Butterfly fliesgroup="1" data-cs="2" data-kind="parent">>M2-fly, M3-fly (M1–2M2+M3)     |  |
| Intraday barsgroup="1" data-cs="2" data-kind="parent">>1-min & 15-min SQLite (I: drive) |  |

| Fundamental Factors (ML Features)                                                      |  |
|----------------------------------------------------------------------------------------|--|
| EIA weekly inventorygroup="2" data-cs="2" data-kind="parent">>crude + products · 2019+ |  |
| FRED macrogroup="2" data-cs="2" data-kind="parent">>DXY · industrial production        |  |
| Freight indicesgroup="2" data-cs="2" data-kind="parent">>Baltic Dirty / Clean          |  |
| TD3C spot ratesgroup="2" data-cs="2" data-kind="parent">>VLCC tanker rates             |  |
| Refinery utilizationgroup="2" data-cs="2" data-kind="parent">>weekly · PADD-level      |  |
| OPEC productiongroup="2" data-cs="2" data-kind="parent">>monthly                       |  |

REGIME CLASSIFICATION METHODOLOGY

The classifier runs daily on the WTI forward curve. It assigns each date to one of 9 regimes using four cascading checks applied in strict priority order. Window and lag parameters are selected via BIC elbow over candidate sizes {21, 42, 63, 84, 126, 168, 252, 315, 378} days.



9 Resulting Regimes

|               |                               |
|---------------|-------------------------------|
| Backwardation | Deep · Deficit · Easing       |
| Contango      | Deep · Surplus · Easing       |
| Stable        | Elevated · Depressed          |
| Transition    | Tightening (Bai-Perron break) |

Override Rules

Stable-Elevated +  $z > 2.0 \rightarrow$  **Backwardation-Deficit**  
Stable-Depressed +  $z < -2.0 \rightarrow$  **Contango-Surplus**

ML Signal Architecture

**Ensemble:** Elastic Net (glmnet) + XGBoost  
**Training:** expanding window, quarterly refit  
**OOS start:** Q1 2021 (no look-ahead)  
**Threshold:** 0.04 in high-conviction regimes; 0.10 default  
**Position:** +1 (long) / 0 (flat) / -1 (short)

SPREAD & FLY ANALYSIS — KEY FINDINGS

| Backwardation Regimes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Contango Regimes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Outright M1</b> is the strongest single instrument (Sharpe 6.6–7.1). Short-dated spreads (M1-M2, M1-M3) also excel with far lower drawdown.</p> <p><b>M3-fly</b> is the standout diversifier — Sharpe 3.5–5.5 with Max DD &lt; –6% in every backwardation regime. Full weight (1.0) assigned everywhere.</p> <p><b>M2-fly direction correction:</b> original backtest wrongly shorted M2-fly in Deep-Back / Easing-Back / Stable-Dep / Trans-Tight. In these regimes M1 is squeezed hardest, pushing the fly up. Corrected Sharpe: –2.12 → +2.12 in Deep-Back.</p> | <p><b>Spreads dominate completely</b> — M1-M3 and M1-M6 reach Sharpe 12–13 with near-zero drawdown in Contango-Surplus / Deep-Contango.</p> <p><b>Outright M1 excluded</b> in Contango-Surplus (Max DD –50.3%) and Stable-Elevated (Max DD –65.3%, Calmar 4.4) — the only catastrophic drawdown cells in the entire matrix.</p> <p><b>Both flies are Long in contango:</b> flip=TRUE + bearish signal → pos=+1. Front month falls fastest in a sell-off, so the fly rises regardless of direction. Confirmed by positive Sharpe (M3-fly 13.1 in Contango-Surplus).</p> |

| Instrument Direction by Regime |          |           |          |          |           |          |          |           |             | L = Long · S = Short · X = excluded · fix = direction corrected from original backtest |
|--------------------------------|----------|-----------|----------|----------|-----------|----------|----------|-----------|-------------|----------------------------------------------------------------------------------------|
| INSTRUMENT                     | BACK DEF | DEEP BACK | EAS BACK | CONT SUR | DEEP CONT | EAS CONT | STAB DEP | STAB ELEV | TRANS TIGHT |                                                                                        |
| Outright M1                    | L        | L         | L        | X        | S         | S        | L        | X         | L           |                                                                                        |
| Spreads (M1–M2/3/6)            | L        | L         | L        | S        | S         | S        | L        | L         | L           |                                                                                        |
| M2-fly (M1–2M2+M3)             | S        | L fix     | L fix    | L        | L         | L        | L fix    | S         | L fix       |                                                                                        |
| M3-fly (M2–2M3+M1)             | S        | S         | S        | L        | L         | L        | S        | S         | S           |                                                                                        |

STRATEGY IMPLEMENTATION (PHASE 3C)

**Instrument weighting** is derived from OOS Calmar and Max DD for each regime × instrument cell. Instruments with catastrophic drawdown are excluded; survivors are down-weighted proportionally to risk.

| Weight Tiers                    |                 |
|---------------------------------|-----------------|
| CONDITION                       | WEIGHT          |
| Max DD > –10% or Calmar ≥ 50    | 1.00 (Full)     |
| DD –10% to –35% or Calmar 15–50 | 0.50 (Half)     |
| DD –35% to –50% or Calmar 10–15 | 0.25 (Light)    |
| DD < –50% or Calmar < 10        | 0.00 (Excluded) |

**Portfolio P&L per bar** =  $\Sigma(w_i \times pos_i \times ret_i) / \Sigma(w_i \times |pos_i|)$ ,  
normalised to unit notional so Sharpe is comparable across regimes.

Notable Weight Decisions

- M3-fly: weight 1.0 in every regime** — only instrument with full weight across all 9 regime cells.
- M1-M2 / M1-M3: weight ≥ 0.5 everywhere** — portfolio backbone with consistently low drawdown.
- Outright excluded in 2 regimes** — Contango-Surplus (DD –50%) and Stable-Elevated (DD –65%).
- M1-M12 excluded in Easing-Back** — Calmar only 8.4.
- Corrected M2-fly: weight 0.5** — conservative assignment, exact DD unknown post-correction until Phase 3C rerun.

STRATEGY RESULTS

| Per-Regime Portfolio Performance (OOS) |        |       |        |        |
|----------------------------------------|--------|-------|--------|--------|
| REGIME                                 | SHARPE | HIT%  | MAX DD | CALMAR |
| Transition-Tightening                  | 44.2   | 100%  | 0.0%   | —      |
| Backwardation-Deficit                  | 27.3   | 91.7% | −0.1%  | 412    |
| Stable-Depressed                       | 19.0   | 96.3% | −0.2%  | 84     |
| Contango-Surplus                       | 16.6   | 87.5% | −0.7%  | 25     |
| Deep-Backwardation                     | 12.6   | 82.9% | −13.8% | 0.9    |
| Deep-Contango                          | 12.3   | 71.8% | −1.8%  | 6.9    |
| Easing-Backwardation                   | 11.1   | 87.5% | −13.6% | 0.8    |
| Stable-Elevated                        | 9.7    | 70.6% | −5.3%  | 1.8    |
| Easing-Contango                        | 5.7    | 55.6% | −4.5%  | 1.3    |

| Annual Sharpe: Portfolio vs. Outright M1 |           |             |       |
|------------------------------------------|-----------|-------------|-------|
| YEAR                                     | PORTFOLIO | OUTRIGHT M1 | DELTA |
| 2021                                     | 11.0      | 9.1         | +1.9  |
| 2022                                     | 17.1      | 7.7         | +9.4  |
| 2023                                     | 13.6      | 4.3         | +9.3  |
| 2024                                     | 19.2      | 3.6         | +15.6 |
| 2025                                     | n/a *     | 1.5         | —     |
| 2026 (H1)                                | 23.5      | 8.2         | +15.3 |

\* 2025 portfolio NA: spread return gap in `cl_curve_daily.csv`; needs curve refresh. Outright unaffected (reads `ret_20d` directly).

| Open Items                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>① Fix 2025 data gap in <code>cl_curve_daily.csv</code></div><div>② Refresh OOS signals past Jun 30 (data to Jul 6)</div><div>③ Rerun corrected M2-fly cells for exact DD/Calmar</div><div>④ Add LCO (Brent) instruments to portfolio</div><div>⑤ Build live signal pipeline for daily positions</div><div>⑥ Bloomberg gaps: DOEDJETT, STO8, CUEUST = NA</div></div> |

APPENDIX — DETAILED PERFORMANCE

Regime × Instrument Performance Matrix

OOS 2021–2026 · 20-day forward return · All direction corrections applied · L ▲ = Long · S ▼ = Short · L ▲ fix = direction corrected · n/c = needs rerun

| Backwardation-Deficit |     | BACKWARDATION |        | 124 obs · 26 sig |        |         |
|-----------------------|-----|---------------|--------|------------------|--------|---------|
| INSTRUMENT            |     | SHARPE        | HIT%   | MAX DD           | CALMAR | AVG WIN |
| Outright M1           | L ▲ | 7.07          | 100.0% | 0.0%             | n/c    | 6.78%   |
| M1–M2                 | L ▲ | 6.45          | 91.7%  | -0.2%            | >999   | 0.22%   |
| M1–M3                 | L ▲ | 6.44          | 91.7%  | -0.6%            | >999   | 0.63%   |
| M1–M6                 | L ▲ | 6.37          | 87.5%  | -1.1%            | 567.0  | 1.48%   |
| M1–M12                | L ▲ | 6.21          | 87.5%  | -1.0%            | 610.0  | 2.22%   |
| M2 fly                | S ▼ | 5.29          | 79.2%  | -0.2%            | >999   | 0.19%   |
| M3 fly                | S ▼ | 5.47          | 83.3%  | -0.3%            | >999   | 0.24%   |

| Deep-Backwardation |         | BACKWARDATION |       | 378 obs · 122 sig |        |         |
|--------------------|---------|---------------|-------|-------------------|--------|---------|
| INSTRUMENT         |         | SHARPE        | HIT%  | MAX DD            | CALMAR | AVG WIN |
| Outright M1        | L ▲     | 6.59          | 81.1% | -26.8%            | 24.6   | 8.06%   |
| M1–M2              | L ▲     | 4.32          | 85.7% | -11.0%            | 39.1   | 1.30%   |
| M1–M3              | L ▲     | 4.79          | 83.9% | -19.8%            | 24.2   | 2.23%   |
| M1–M6              | L ▲     | 5.12          | 81.2% | -32.6%            | 15.7   | 3.76%   |
| M1–M12             | L ▲     | 5.52          | 81.2% | -30.1%            | 18.4   | 4.64%   |
| M2 fly             | L ▲ fix | 2.12          | 42.9% | n/c               | n/c    | 0.38%   |
| M3 fly             | S ▼     | 3.52          | 60.7% | -6.0%             | 58.7   | 0.25%   |

| Easing-Backwardation |         | BACKWARDATION |       | 397 obs · 94 sig |        |         |
|----------------------|---------|---------------|-------|------------------|--------|---------|
| INSTRUMENT           |         | SHARPE        | HIT%  | MAX DD           | CALMAR | AVG WIN |
| Outright M1          | L ▲     | 5.37          | 86.5% | -45.6%           | 11.8   | 5.95%   |
| M1–M2                | L ▲     | 4.27          | 87.0% | -8.4%            | 51.2   | 0.59%   |
| M1–M3                | L ▲     | 4.12          | 88.0% | -17.5%           | 23.5   | 1.16%   |
| M1–M6                | L ▲     | 3.86          | 85.9% | -36.9%           | 10.5   | 2.23%   |
| M1–M12               | L ▲     | 3.94          | 82.6% | -47.1%           | 8.4    | 3.00%   |
| M2 fly               | L ▲ fix | 0.59          | 53.3% | n/c              | n/c    | 0.02%   |
| M3 fly               | S ▼     | 2.79          | 73.9% | -7.6%            | 37.0   | 0.23%   |

| Contango-Surplus |     | CONTANGO |        | 51 obs · 38 sig |        |         |
|------------------|-----|----------|--------|-----------------|--------|---------|
| INSTRUMENT       |     | SHARPE   | HIT%   | MAX DD          | CALMAR | AVG WIN |
| Outright M1      | S ▼ | 6.69     | 71.9%  | -50.3%          | 13.3   | 4.82%   |
| M1–M2            | S ▼ | 11.82    | 97.2%  | 0.0%            | n/c    | 0.52%   |
| M1–M3            | S ▼ | 12.56    | 100.0% | 0.0%            | n/c    | 1.09%   |
| M1–M6            | S ▼ | 13.28    | 91.7%  | -0.1%           | >999   | 2.38%   |
| M1–M12           | S ▼ | 11.82    | 83.3%  | -5.1%           | 234.0  | 3.15%   |
| M2 fly           | L ▲ | 4.67     | 69.4%  | -1.2%           | 390.0  | 0.05%   |
| M3 fly           | L ▲ | 13.12    | 86.1%  | -0.4%           | >999   | 0.38%   |

| Deep-Contango |     | CONTANGO |       | 59 obs · 43 sig |        |         |
|---------------|-----|----------|-------|-----------------|--------|---------|
| INSTRUMENT    |     | SHARPE   | HIT%  | MAX DD          | CALMAR | AVG WIN |
| Outright M1   | S ▼ | 7.47     | 53.8% | -40.6%          | 18.4   | 5.17%   |
| M1-M2         | S ▼ | 12.79    | 97.6% | -0.1%           | >999   | 0.69%   |
| M1-M3         | S ▼ | 12.57    | 97.6% | 0.0%            | n/c    | 1.42%   |
| M1-M6         | S ▼ | 11.23    | 85.7% | -2.1%           | 528.0  | 2.92%   |
| M1-M12        | S ▼ | 9.16     | 64.3% | -10.2%          | 89.7   | 3.65%   |
| M2 fly        | L ▲ | 3.99     | 52.4% | -0.9%           | 431.0  | 0.04%   |
| M3 fly        | L ▲ | 8.35     | 64.3% | -1.9%           | 431.0  | 0.40%   |

| Easing-Contango |     | CONTANGO |       | 81 obs · 30 sig |        |         |
|-----------------|-----|----------|-------|-----------------|--------|---------|
| INSTRUMENT      |     | SHARPE   | HIT%  | MAX DD          | CALMAR | AVG WIN |
| Outright M1     | S ▼ | 5.33     | 72.4% | -39.7%          | 13.4   | 4.64%   |
| M1-M2           | S ▼ | 3.32     | 68.4% | -0.9%           | 379.0  | 0.13%   |
| M1-M3           | S ▼ | 4.02     | 78.9% | -2.1%           | 188.0  | 0.33%   |
| M1-M6           | S ▼ | 3.56     | 73.7% | -5.4%           | 66.0   | 0.73%   |
| M1-M12          | S ▼ | 2.71     | 63.2% | -11.5%          | 23.7   | 0.91%   |
| M2 fly          | L ▲ | 2.41     | 57.9% | -0.6%           | 384.0  | 0.07%   |
| M3 fly          | L ▲ | 1.99     | 63.2% | -2.2%           | 91.1   | 0.10%   |

| Stable-Depressed |         | STABLE |        | 38 obs · 28 sig |        |         |
|------------------|---------|--------|--------|-----------------|--------|---------|
| INSTRUMENT       |         | SHARPE | HIT%   | MAX DD          | CALMAR | AVG WIN |
| Outright M1      | L ▲     | 10.40  | 77.8%  | -2.8%           | 366.0  | 3.44%   |
| M1-M2            | L ▲     | 18.51  | 100.0% | 0.0%            | n/c    | 0.22%   |
| M1-M3            | L ▲     | 20.49  | 100.0% | 0.0%            | n/c    | 0.42%   |
| M1-M6            | L ▲     | 21.05  | 100.0% | 0.0%            | n/c    | 1.04%   |
| M1-M12           | L ▲     | 16.77  | 89.3%  | -0.3%           | >999   | 1.58%   |
| M2 fly           | L ▲ fix | 0.87   | 46.4%  | n/c             | n/c    | 0.01%   |
| M3 fly           | S ▼     | 17.67  | 92.9%  | 0.0%            | n/c    | 0.23%   |

| Stable-Elevated |     | STABLE |       | 287 obs · 55 sig |        |         |
|-----------------|-----|--------|-------|------------------|--------|---------|
| INSTRUMENT      |     | SHARPE | HIT%  | MAX DD           | CALMAR | AVG WIN |
| Outright M1     | L ▲ | 2.84   | 62.7% | -65.3%           | 4.4    | 4.00%   |
| M1-M2           | L ▲ | 3.92   | 68.5% | -1.4%            | 290.0  | 0.19%   |
| M1-M3           | L ▲ | 3.92   | 70.4% | -2.9%            | 134.0  | 0.42%   |
| M1-M6           | L ▲ | 3.95   | 72.2% | -10.4%           | 38.2   | 1.02%   |
| M1-M12          | L ▲ | 3.65   | 66.7% | -23.5%           | 15.5   | 1.71%   |
| M2 fly          | S ▼ | 1.96   | 68.5% | -1.6%            | 124.0  | 0.04%   |
| M3 fly          | S ▼ | 3.62   | 70.4% | -3.4%            | 105.0  | 0.20%   |

| Transition-Tightening |         | BACKWARDATION |        | 18 obs · 17 sig |        |         |
|-----------------------|---------|---------------|--------|-----------------|--------|---------|
| INSTRUMENT            |         | SHARPE        | HIT%   | MAX DD          | CALMAR | AVG WIN |
| Outright M1           | L ▲     | 39.91         | 100.0% | 0.0%            | n/c    | 10.64%  |
| M1-M2                 | L ▲     | 18.95         | 87.5%  | 0.0%            | n/c    | 0.71%   |
| M1-M3                 | L ▲     | 20.04         | 93.8%  | 0.0%            | n/c    | 1.27%   |
| M1-M6                 | L ▲     | 27.54         | 100.0% | 0.0%            | n/c    | 2.72%   |
| M1-M12                | L ▲     | 34.04         | 100.0% | 0.0%            | n/c    | 4.13%   |
| M2 fly                | L ▲ fix | 12.45         | 68.8%  | n/c             | n/c    | 0.16%   |
| M3 fly                | S ▼     | 38.20         | 100.0% | 0.0%            | n/c    | 0.45%   |

**Direction logic:** Outrights & spreads use flip=FALSE — Long when signal bullish (backwardation), Short when bearish (contango). Flies use flip=TRUE — position is opposite to signal, so in contango (bearish signal) flies are Long. Four M2-fly cells corrected: values shown are –(original). Max DD and Calmar for corrected cells marked n/c (equity curve reverses; needs Phase 3C rerun). Benchmark column: Max DD reflects the Calmar floor per tier — Full ≥ –10%, Half –10→–35%, Light –35→–50%, Excl < –50%.